

Week 6 Deliverable: Statistical Analysis Results and Report Draft

1. Linear Regression: Hours Spent in Weekly Study and Math Score

- Coefficient: +2.31 (every hour of self-studying math leads to an improvement of math scores by ~2.3 points)
- R-squared: 0.49 (49 percent of the variation in the scores has been accounted for)
- p-value: < 0.05

Conclusion: The higher the self-study, the higher the math performance.

2. ANOVA: Extracurricular Activities Vs Math Score

- Null Hypothesis (H_0): There is no difference between mean math scores
- p-value: 0.03 ✓ Reject H_0

Conclusion: A student engaged in extracurricular activities gets better marks in math.

3. An Identifier of High Performers: Logistic Regression

- Definition: High Performer = >70 = Average Score
- Features Used Weekly Study Hours, Absence Days
- The accuracy is approximately 82%.

Conclusion: The weekly study and attendance are good predictors of student success.

4. Other Insights

- English and Biology scores have shown a high rate of correlation (visually on heatmap)
- There are no significant violations of the model as shown in the residual plots

Draft Structure of Report

Title:

A Statistical Study of the Performance at the Academic Level

I. Introduction

- The need to know the issues influencing student performance
- Purpose of the research

II. Methodology

- Sling Academy dataset
- Preprocessing: outliers, missing data, and feature engineering

III. Statistical Analysis

- Descriptive statistics
- Statistical analysis (Inferential) (ANOVA, Linear Regression, Logistic Regression)
- Diagnostic tests

IV. Findings and Observations

- Key findings summary
- Model results interpretation

V. Recommendations and Conclusion

- Overview of data-based best practices to raise the performance of students

VI. Appendices

- Examples of codes
- Visualizations (scatter, heatmaps, pie charts, bar plots, etc)